

# Graphs, Networks, Incidence Matrices

Turn a network of prices into a matrix.  
Loops become the **left null space**, and a  
loop that doesn't add to zero is an **arbitrage**.

$$\text{nodes} - \text{edges} + \text{loops} = 1$$

$$A^T y = 0$$

MIT 18.06, Lecture 12

# Contracts and spreads

**Nodes:** Nifty spot, Oct, Nov, Dec futures (20,000 / 20,100 / 20,250 / 20,400).

**Edges:** quoted spreads, e.g. Oct – spot = 100.

**Incidence matrix:** one row per edge,  $-1$  at the start node,  $+1$  at the end node. Then

$Ax$  = the spread on every edge.

# Only relative prices matter

$Ax = 0 \Rightarrow$  all prices equal:  $N(A) = c(1, 1, 1, 1)$ .

Add 50 points to every price and **no spread changes**.

## GROUND A NODE

Fix spot = 20,000 (the numeraire) and the spreads decide every other price. Rank =  $n - 1$ .

# Positions that cancel

Hold  $y_i$  lots of spread  $i$ . Then  $A^T y$  = net position in each contract.

$A^T y = 0$  (Kirchhoff's law): flat in every contract.  
Its solutions are **loops**:

$$y = (1, 1, -1, 0, 0)$$

buy spot→Oct, buy Oct→Nov, sell spot→Nov: net zero in every contract.

# Euler's formula

Loops =  $\dim N(A^T) = m - (n - 1)$ . So

$$\text{nodes} - \text{edges} + \text{loops} = 1$$

4 contracts, 5 spreads:  $4 - 5 + 2 = 1 \checkmark$ . Two independent loops.

# No arbitrage = every loop sums to 0

Quotes come from real prices  $\Leftrightarrow e \in C(A) \Leftrightarrow y^T e = 0$  for every loop (Lesson 10's test).

Consistent:  $100 + 150 - 250 = 0 \checkmark$

Mispriced spot  $\rightarrow$  Nov at 240:  $100 + 150 - 240 = 10 \times$

# Trade the loop

Buy the direct spot→Nov spread at 240, sell spot→Oct and Oct→Nov (worth 250).

## RESULT

Net 0 lots in every contract ( $A^T y = 0$ ), 10 points locked in.

Currency triangles work the same way with logs:  
 $83 \times 1.08 = 89.64$ , so a ₹90/€ quote is 0.4% off.

RECAP

# Four subspaces of a network

| Space    | Dim         | Meaning                   |
|----------|-------------|---------------------------|
| $N(A)$   | 1           | shift all prices          |
| $C(A)$   | $n - 1$     | consistent quotes         |
| $C(A^T)$ | $n - 1$     | positions from spreads    |
| $N(A^T)$ | $m - n + 1$ | loops: the arbitrage test |

QUANT FROM FIRST PRINCIPLES

# The full interactive lesson is free

A spread checker, a network lab that finds loops and arbitrage, and worked currency and lot-size examples:

**[learn.krisarj.com](https://learn.krisarj.com)**

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